Experimental Techniques

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

1. An experiment is a designed argument
2. Differential photometry
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1. An experiment is a designed argument

A good experiment begins with a question that can be answered by a measurement. What is the variable? What is held constant? What is the control? What calibration is needed? What uncertainty will limit the answer? The method should be written before observing, so the result is not chosen after the fact.

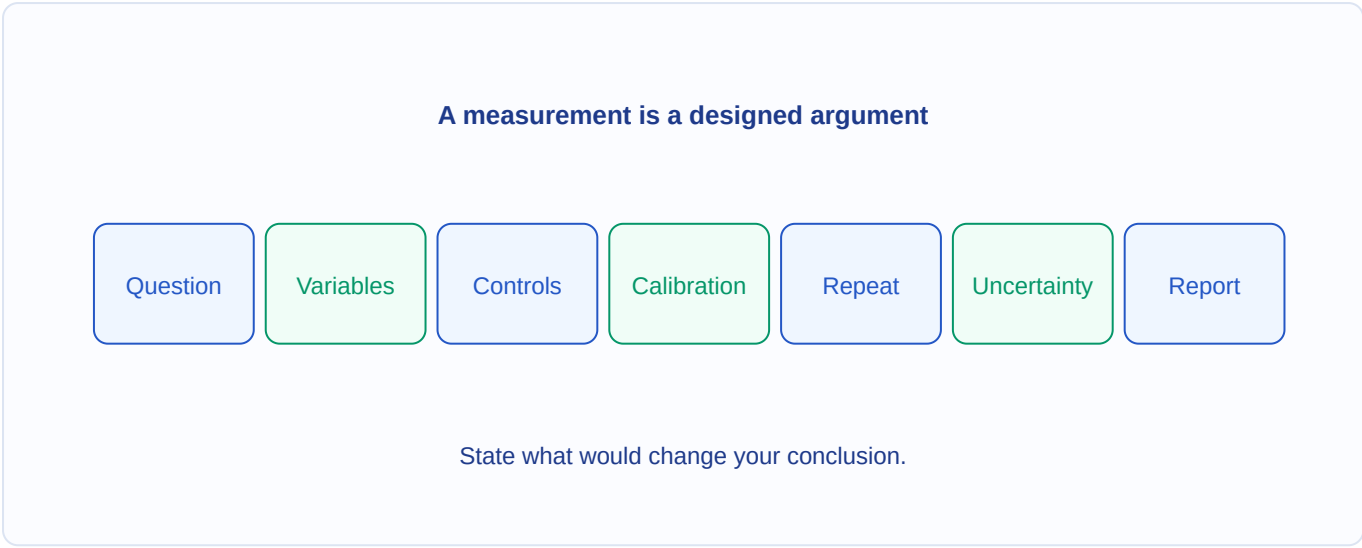


Figure 2. The final report includes method, data, uncertainty and limitations.

2. Differential photometry

Differential photometry measures a target relative to comparison stars in the same field. If the target and comparison share the same transparency, airmass and instrumental trends, the ratio cancels common variations. In magnitudes,

$$\Delta m = m_{\mathrm{target}} - m_{\mathrm{comparison}} = -2.5 \log_{10} \left(\frac{F_{\mathrm{target}}}{F_{\mathrm{comparison}}} \right) . \tag{1}$$

A stable comparison star is essential. Choose stars of similar brightness and colour, away from crowding, and check that they are not variable. Multiple comparison stars improve reliability. The differential magnitude is relative to the chosen comparison ensemble; a calibrated zero point is needed to turn it into an absolute magnitude.

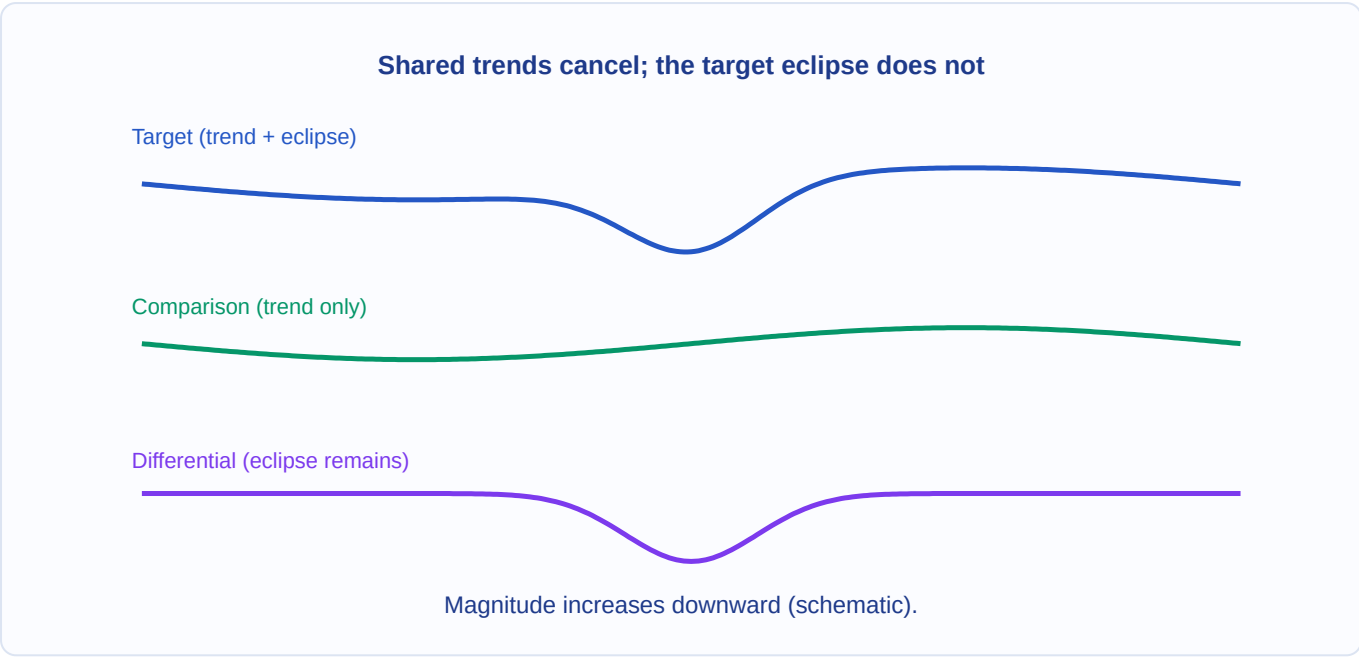


Figure 1. The comparison must be stable and close enough to share the same systematics.

3. Calibration and controls

Bias, dark and flat frames remove instrumental signatures. Standard stars or catalogue magnitudes set the zero point. For time-series photometry, keep exposure time, filter, focus and aperture consistent. If the target changes airmass, either correct for extinction or use comparisons at similar airmass.

A control is a measurement that should not change. If the comparison star shows the same “eclipse” as the target, the signal is probably instrumental or atmospheric, not astrophysical.

4. Cadence, duration and phase

Time-series work needs enough points to sample the event. For an eclipse, observe before, during and after. The cadence should resolve ingress and egress, not just the minimum. For periodic variables, cover multiple cycles if possible and record accurate times. Phase is

$$\phi = \frac{t - T_0}{P} \bmod 1. \tag{2}$$

A timing error shifts phase; a period error accumulates over many cycles.

5. Noise, systematics and significance

Photon noise sets a floor, but systematics often dominate: flat-field errors, colour terms, focus drift, tracking changes, comparison variability and aperture losses. Estimate the scatter from out-of-eclipse data. A claimed eclipse depth should be several times larger than the residual scatter and should **not** appear in independent stable comparison stars; a shared dip is evidence for an instrumental or atmospheric systematic.

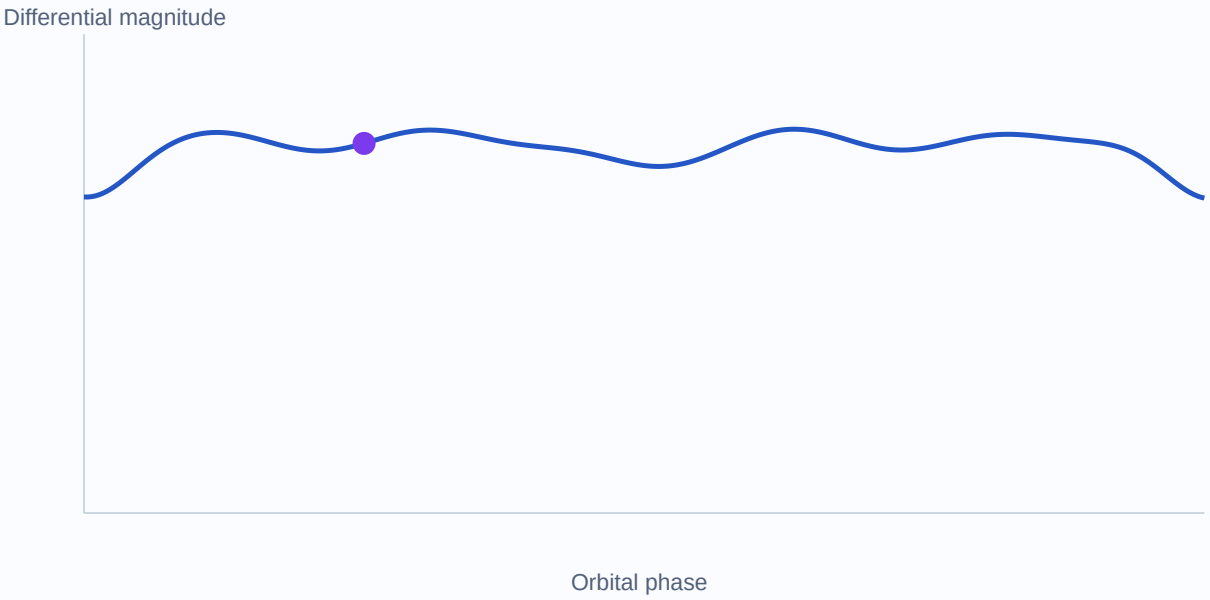
INTERACTIVE EXPERIMENT

Differential photometry cancels shared trends.

A synthetic eclipsing system is compared with a stable comparison star. The differential magnitude removes a common transparency trend, leaving the eclipse and deterministic synthetic ripple.

Differential magnitude	Model magnitude	Residual
0.0068 mag	**0.0000 mag**	**0.0068 mag**

Input depth scale
0.100 mag



Differential photometry removes shared transparency trends. A constant comparison-star offset shifts the zero point but not the eclipse shape. The ripple is deterministic synthetic scatter for teaching.

Try this. Move through phase 0 and 0.5. Where are the eclipses? Increase the comparison offset: does the differential curve shift?

Snapshot of the experiment. Open the HTML edition to change the controls.

6. Reporting an experimental result

Report the question, method, calibration, data, uncertainty and limitations. Show the light curve, the comparison stars, the residual scatter and the fitted depth or period. State what would change the conclusion. A negative result is still a result if the sensitivity is quantified.

A reliable method. Define the question, choose controls, calibrate, repeat, and quantify the noise. A signal is only as strong as the systematic errors you have ruled out.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Differential magnitude. A target has flux 100 units and a comparison has flux 200 units. Find the differential magnitude.

Worked solution →

[1] Problem 2 — Eclipse depth. A star's flux drops by 10% during eclipse. What is the approximate magnitude depth?

Worked solution →

[2] Problem 3 — Comparison-star choice. List three properties of a good comparison star for differential photometry.

Worked solution →

[2] Problem 4 — Common trend. Transparency drops by 20% for all stars in the field. What happens to the differential magnitude of a non-variable target? Why?

Worked solution →

[2] Problem 5 — Phase. A variable has period 2.5 days and $T_0 = 0$. What is the phase at $t = 6.25$ days?

Worked solution →

[1] Problem 6 — Calibration order. Why are bias, dark and flat frames taken before scientific analysis? What does each remove?

Worked solution →

[2] Problem 7 — Significance. An eclipse depth is 0.03 mag and the out-of-eclipse scatter is 0.01 mag. Is the detection convincing? What else should be checked?

Worked solution →

[2] Problem 8 — Cadence. An eclipse lasts 90 minutes. You want 15 points during the eclipse. What exposure-plus-readout cadence is needed? Why observe outside eclipse too?

Worked solution →

[3] Problem 9 — False signal. The target and comparison star both show a 0.05 mag dip at the same time. What is the likely cause? How would you test it?

Worked solution →

[3] Problem 10 — Design an experiment. Design a differential-photometry experiment to measure a 0.05 mag eclipse of a magnitude 11 star. Include comparison stars, calibration, cadence and one systematic to control.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.